NORMALIZED BASE CHANGE ALONG A SEPARABLE RESIDUE EXTENSION IS SMOOTH AT GENERIC FIBRE POINTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19412-19478, immediately following the statement at 19400-19410.

Editorial note. The proof opens with 'This is proved in exactly the same manner as [the general theorem]' and then reproduces the argument for the separable case; the reproduced argument is inside the span.

Editorial note. Nine cross-chapter pointers (More on Algebra, Algebra, Morphisms, Varieties, Decent Spaces) are recorded in assembly.external_references as navigation pointers.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 19467 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Normalized base change of the general theorem. chapter theorem theorem-normalized-base-change-with-reduced-fibre, cited at proof line 19412 onward. Supplies the framework (normalization T, normalized base change Y) whose smoothness locus is refined here.
- Separable field extensions are smooth. Algebra, lemma algebra-lemma-separable-smooth, cited at proof line 19476. Converts separability of the residue field extension at a point into smoothness of the fibre there.
- Solutions of an essentially finite type problem go up. More on Algebra, lemmas more-algebra-lemma-epp-essentially-finite-type-separable, more-algebra-lemma-solution-goes-up and proposition more-algebra-proposition-epp-essentially-finite-type, cited throughout the proof. Provides the algebra result that a separable solution to the essentially finite type problem exists after a finite separable extension.
- Where the fibres are smooth and generically reduced. Morphisms, lemma morphisms-lemma-set-points-where-fibres-smooth and Varieties, lemma varieties-lemma-generic-points-geometrically-reduced, cited in the proof. Identifies the loci of points where the fibres are smooth and controls generic points of the fibres.

Source reference. Original reference 033C: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TV: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 5–5. Navigation pointer only: the

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Source reference. Original equation 09IK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19260–19260. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 054X: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6524–6524. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BRP: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GLR: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09II: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01V5: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02V4: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 09IL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19282–19282. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04KS: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 056V: focused-seeds/geometry/S9-morph-normalized-base-change-separable/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Variant with separable extension)**.** *Let A be a Dedekind ring with fraction field K. Let X be a scheme flat and of finite type over A. Assume A is a Nagata ring and that for every generic point η of an irreducible component of X the field extension $\kappa(\eta)/K$ is separable. Then there exists a finite separable extension L/K such that the normalized base change Y is smooth over* $\mathrm{Spec}(B)$ *at all generic points of all fibres.*

Proof. This is proved in exactly the same manner as Theorem 09IL with a few minor modifications. The most important change is to use More on Algebra, Lemma 0BRP instead of More on Algebra, Proposition 09II. During the proof we will repeatedly use that formation of the set of points where a (flat, finitely presented) morphism like $X \to \mathrm{Spec}(A)$ is smooth commutes with base change, see Morphisms, Lemma 02V4.

Since X is flat over A every generic point η of X maps to the generic point of $\mathrm{Spec}(A)$. After replacing X by its reduction we may assume X is reduced. In this case X_K is geometrically reduced over K by Varieties, Lemma 04KS. Hence $X_K \to \mathrm{Spec}(K)$ is smooth on a dense open by Varieties, Lemma 056V. Thus the smooth locus $U \subset X$ of the morphism $X \to \mathrm{Spec}(A)$ is open (by Morphisms, Definition 01V5) and is dense in the generic fibre. This reduces us to the situation of the following paragraph.

Assume X is normal and the smooth locus $U \subset X$ of $X \to \mathrm{Spec}(A)$ is dense in the generic fibre. This implies that U is dense in all but finitely many fibres, see Lemma 054X. Let $x_1, \ldots, x_r \in X \setminus U$ be the finitely many generic points of irreducible components of $X \setminus U$ which are moreover generic points of irreducible components of fibres of $X \to \mathrm{Spec}(A)$. Set $\mathcal{O}_i = \mathcal{O}_{X,x_i}$. Observe that the fraction field of $\mathcal{O}_i$ is the residue field of a generic point of X. Let A_i be the localization of A at the maximal ideal corresponding to the image of x_i in $\mathrm{Spec}(A)$. We may apply More on Algebra, Lemma 0BRP and we find finite separable extensions K_i/K which are solutions for $A_i \to \mathcal{O}_i$. Let L/K be a finite separable extension dominating all of the extensions K_i/K. Then L/K is still a solution for $A_i \to \mathcal{O}_i$ by More on Algebra, Lemma 0GLR.

Consider the diagram (09IK) with the extension L/K we just produced. Note that $U_B \subset X_B$ is smooth over B, hence normal (for example use Algebra, Lemma 033C). Thus $Y \to X_B$ is an isomorphism over U_B. Let $y \in Y$ be a generic point of an irreducible component of a fibre of $Y \to \mathrm{Spec}(B)$ lying over the maximal ideal $\mathfrak{m} \subset B$. Assume that $y \notin U_B$. Then y maps to one of the points x_i. It follows that $\mathcal{O}_{Y,y}$ is a local ring of the integral closure of $\mathcal{O}_i$ in $R(X) \otimes_K L$ (details omitted). Hence because L/K is a solution for $A_i \to \mathcal{O}_i$ we see that $B_{\mathfrak{m}} \to \mathcal{O}_{Y,y}$ is formally smooth (this is the definition of being a "solution"). In other words, $\mathfrak{m}\mathcal{O}_{Y,y} = \mathfrak{m}_y$ and the residue field extension is separable. Hence the local ring of the fibre at y is $\kappa(y)$. This implies the fibre is smooth over $\kappa(\mathfrak{m})$ at y for example by Algebra, Lemma 00TV. This finishes the proof. □